

- 1) **A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system.**

```
# Scan the target IP address for common open ports  
nmap 192.168.1.10
```

```
# -sV enables service version detection  
# It identifies the version of services running on open ports  
nmap -sV 192.168.1.10
```

```
# sudo runs the command with administrator privileges  
# -O detects the operating system of the target  
sudo nmap -O 192.168.1.10
```

```
# sudo gives administrator privileges  
# -A enables aggressive scan  
# It performs OS detection, service version detection,  
# default script scanning, and traceroute  
sudo nmap -A 192.168.1.10
```

- 2) **A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.**

```
# Download the latest GoPhish package from GitHub
```

```
wget https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
```

```
# Extract the downloaded ZIP file
```

```
unzip gophish-v0.12.1-linux-64bit.zip
```

Give execute permission to the GoPhish application

```
chmod +x gophish
```

Run the GoPhish phishing simulation server

sudo provides administrator privileges

```
sudo ./gophish
```

3) A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure.

Query DNS information for the domain

```
dig example.com
```

Display the Mail Exchange (MX) records

These records show the mail servers responsible for the domain

```
dig MX example.com
```

Display the Name Server (NS) records

These records identify the authoritative DNS servers

```
dig NS example.com
```

Attempt a DNS Zone Transfer (AXFR)

Replace example.com with the target domain

Replace ns1.example.com with the authoritative DNS server

Only test on systems you own or have permission to assess

```
dig AXFR example.com @ns1.example.com
```